

2. 设 $P(X=k) = \frac{C}{k(k+1)}$, 且 $\sum_{n=1}^{\infty} \frac{C}{n(n+1)} = 1$ 求: $C=1$

$\therefore P(X=k) = \frac{1}{k(k+1)}$

3. $P(X=1) = \frac{C_3^1}{C_4^2} = \frac{1}{2}$ $P(X=2) = \frac{C_2^1}{C_4^2} = \frac{1}{3}$

$P(X=3) = \frac{C_1^1}{C_4^2} = \frac{1}{6}$

X	1	2	3
P	$\frac{1}{2}$	$\frac{1}{3}$	$\frac{1}{6}$

5. $P(X=i, Y=i) = \frac{1}{6^n}$

$P(X=i, Y=j) \quad (i \neq j)$
 $= \frac{(j-i+1)^n - 2(j-i)^n + (j-i-1)^n}{6^n}$

$\therefore P(X=i) = \frac{(7-i)^n - (6-i)^n}{6^n}$

$P(Y=i) = \frac{i^n - (i-1)^n}{6^n}$

$P(X=2, Y=5) = \frac{4^n - 2 \times 3^n + 2^n}{6^n}$

6. $P(X=0) = \frac{3}{4}$

$P(X=1) = \frac{1}{4} \times \frac{9}{11} = \frac{9}{44}$

$P(X=2) = \frac{1}{4} \times \frac{2}{11} \times \frac{9}{10} = \frac{9}{220}$

$P(X=3) = 1 - (P(X=1) + P(X=2) + P(X=3)) = \frac{1}{220}$

$$7. P(X=2, Y=1) = P(X=2, Y=3) = 0 \quad P(X=2, Y=4) = \frac{2 \times \frac{2}{3}}{3^3} = \frac{6}{27}$$

$$P(X=3, Y=2) = P(X=3, Y=3) = 0, \quad P(X=3, Y=1) = \frac{1}{3^3} = \frac{1}{27}$$

$$P(X=0, Y=1) = \frac{2}{27} \quad P(X=0, Y=2) = \frac{6}{27} \quad P(X=0, Y=3) = 0$$

$$P(X=1, Y=1) = 0 \quad P(X=1, Y=2) = \frac{6}{27} \quad P(X=1, Y=3) = \frac{6}{27}$$

$$\therefore$$

$X \backslash Y$	1	2	3	P_i^X
0	$\frac{2}{27}$	$\frac{6}{27}$	0	$\frac{8}{27}$
1	0	$\frac{6}{27}$	$\frac{6}{27}$	$\frac{12}{27}$
2	0	$\frac{6}{27}$	0	$\frac{6}{27}$
3	$\frac{1}{27}$	0	0	$\frac{1}{27}$
P_j^Y	$\frac{1}{9}$	$\frac{2}{3}$	$\frac{2}{9}$	1

9. 设 X_i 为第 i 次取出卡片的号码.

显然, 放回不放回, $P(X_i = k) = \frac{1}{n}$ 不变 $E X_i = \frac{n+1}{2}$

$$\therefore E X = E(X_1 + X_2 + \dots + X_k) = E X_1 + E X_2 + \dots + E X_k = \frac{k(n+1)}{2}$$

$$12. p = \frac{3^3 + 3 \times 3^2}{6^3} = \frac{1}{2}$$

设试验所需次数为 X , 显然 $X \sim \text{Geol}(p)$

$$\therefore E X = 2, \text{ 即平均次数为 } 2$$



$$\begin{aligned}
 16. EY &= 1 \cdot C_n^{k0} \cdot p^0 \cdot q^{n+0} + (-1) \cdot C_n^1 p^1 q^{n-1} + \dots \\
 &= \sum_{k=0}^n C_n^k (-1)^k q^{n-k} \\
 &= (-p+q)^n = (1-2p)^n
 \end{aligned}$$

$$\begin{aligned}
 19. (a) E X &= \sum_{k=1}^{\infty} k P(X=k) = \sum_{k=1}^{\infty} \sum_{n=k}^{\infty} P(X=n) \\
 &= \sum_{n=1}^{\infty} \sum_{k=n}^{\infty} P(X=n) = \sum_{n=1}^{\infty} P(X \geq n)
 \end{aligned}$$

20. $X \geq k$, 意味着前 $k-1$ 局一直是两人轮流获胜.

$$P(X \geq 1) = 1$$

$$P(X \geq 2k+1) = 2p^k q^k$$

$$P(X \geq 2k) = p^k q^{k-1} + p^{k-1} q^k = (pq)^{k-1} \quad k=1, 2, \dots$$

$$EX = \sum_{n=1}^{\infty} P(X \geq n) = \frac{2+pq}{1-pq}$$

23. (a)

X	1	2	3
P	$\frac{1}{4}$	$\frac{2}{3}$	$\frac{2}{9}$

Y	0	1	2	3
P	$\frac{8}{27}$	$\frac{4}{9}$	$\frac{2}{9}$	$\frac{1}{27}$

(b) 不独立

(c) ~~P(X=0)~~

Y X=0	0	1	2	3
P	$\frac{2}{3}$	0	0	$\frac{1}{3}$

X Y=0	1	2	3
P	$\frac{1}{4}$	$\frac{3}{4}$	0

$$(d) P(X=3 | Y=2) = 0$$

$$P(Y=2 | X=3) = 0$$

补充题:



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1. 即前 $m+n-1$ 次实验, 失败了 $m-1$ 次, 成功了 n 次, 且第 $m+n$ 次失败

$$p = C_{m+n-1}^{m-1} q^{m-1} p^n, q = C_{m+n-1}^m q^m p^n$$

$$2. P(X=k) = \frac{k^n - (k-1)^n}{N^n}$$

$$E X = \sum_{k=1}^N k P(X=k) = \sum_{k=1}^N k \frac{k^n - (k-1)^n}{N^n} = \frac{1}{N^n} (N^{n+1} - \sum_{k=1}^{N-1} k^n)$$

$$= N - \frac{(1-k^n)}{1-k} \frac{1}{N^n} \sum_{k=1}^{N-1} k^n$$

故 $\sum_{k=1}^{N-1} k^n$ 可由递推得到.